

Display Style in Math Mode

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July 2, 2026

Introduction

The canonical example is taken from page 142 of the *TeXBook*, although we've replaced $\$$ by L^AT_EX's preferred `\[` and `\]`:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}}}$$

The default typesetting style can be amended by using the `\displaystyle` command:

$$a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4}}}}$$

Here's another example which demonstrates the effect of `\textstyle`, `\scriptstyle`, and `\scriptscriptstyle`:

$$\begin{aligned} f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\ f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\ f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\ f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \end{aligned}$$

Depending on the value of x , the equation

$$f(x) = \sum_{i=0}^n \frac{a_i}{1+x}$$

may diverge or converge.

Inline maths elements can be set with a different style: $f(x) = \frac{1}{1+x}$.

The same is true for display math material:

$$\begin{aligned}
 f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\
 f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\
 f(x) &= \sum_{i=0}^n \frac{a_i}{1+x} \\
 f(x) &= \sum_{i=0}^n \frac{a_i}{1+x}
 \end{aligned}$$